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Task 1:

TT:

A1	**A0**	**B1**	**B0**	**L (A<B)**	**E (A=B)**	**G (A>B)**
0	0	0	0	0	1	0
0	0	0	1	1	0	0
0	0	1	0	1	0	0
0	0	1	1	1	0	0
0	1	0	0	0	0	1
0	1	0	1	0	1	0
0	1	1	0	1	0	0
0	1	1	1	1	0	0
1	0	0	0	0	0	1
1	0	0	1	0	0	1

A1	A0	B1	B0	L (A<B)	E (A=B)	G (A>B)
1	0	1	0	0	1	0
1	0	1	1	1	0	0
1	1	0	0	0	0	1
1	1	0	1	0	0	1
1	1	1	0	0	0	1
1	1	1	1	0	1	0

Expression:

Simplified expression

Equal - Both equal

$$\boxed{E = (A_1 \odot B_1) \cdot (A_0 \odot B_0)}$$

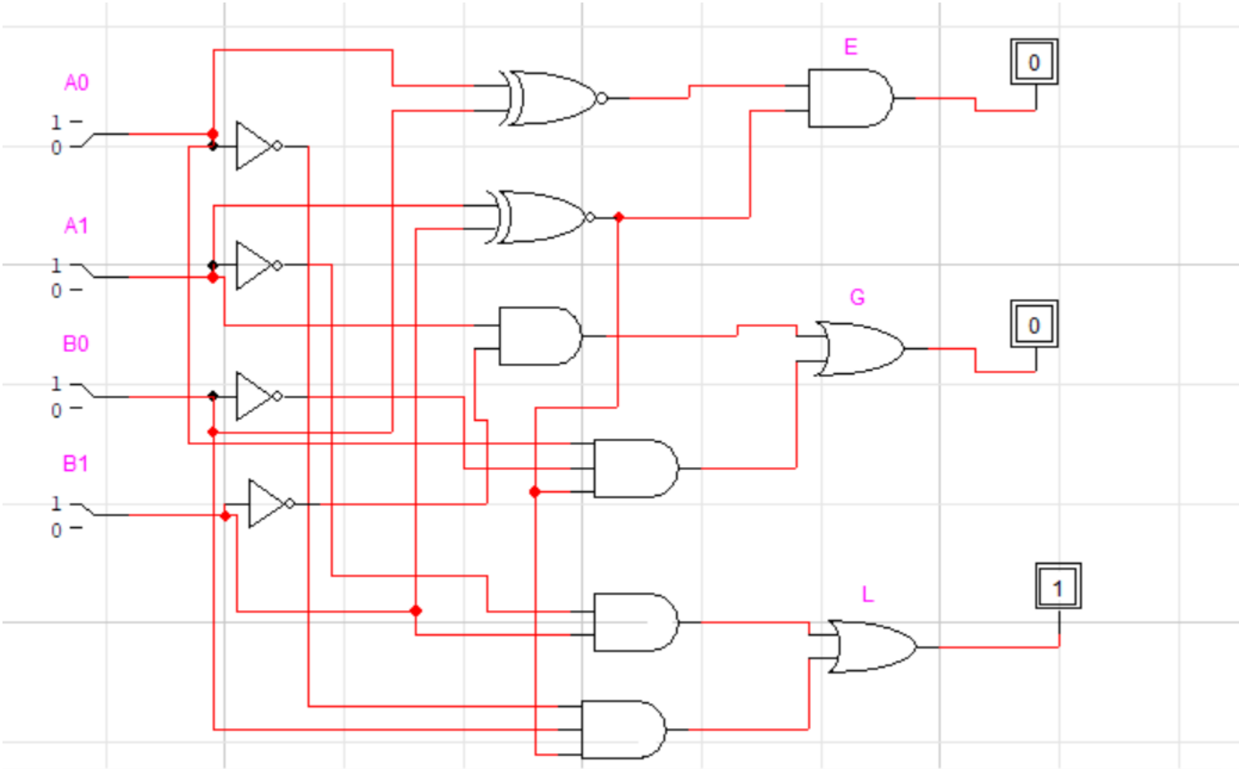
Greater $A_1 > B_1$ or $A_1 = B_1$ and $A_0 > B_0$

$$\boxed{G = A_1 \bar{B} + (A_1 \odot B_1)\, A_0 \bar{B}_0}$$

Less than $A_1 < B_1$ or $(A_1 = B_1$ and $A_0 < B_0)$

$$\boxed{L = \bar{A}_1 B_1 + (A_1 \odot B_1)\, \bar{A}_0 B_0}$$

Circuit Diagram:



1-bit comparator:

Circuit implementation